

Krish Ruparel

Arlington, Texas 76013 • krishruparel.career@gmail.com • (682)-392-0214 • [LinkedIn](#) • [GitHub](#) • [Portfolio](#)

Education

University of Texas, Arlington

Master of Science - MS, Computer Science

Coursework: Advanced Algorithms, Data Modelling, Data Mining, Artificial Intelligence, Computer Vision

Aug 2025 - May 2027

GPA: 3.84/4.0

University of Pune

Bachelor of Engineering - BE, Computer Engineering

Coursework: Data Structures, Database Management, Software Engineering, Machine Learning, Cloud Computing and Big Data

Aug 2021 - May 2025

GPA: 3.4/4.0

Skills

- **Programming Skills:** Python, R, C++, Java, JavaScript, SQL, CSS, HTML
- **Data Visualization:** Tableau, Power BI, SAS, AWS QuickSight, SAP Business Intelligence, MATLAB
- **Database Technology:** PostgreSQL, MySQL, MongoDB, DynamoDB, Couchbase, Hadoop, Apache Spark, Data Warehousing, Data Lakes
- **Machine Learning:** Numpy, Pandas, Matplotlib, Pyspark, Seaborn, Scikit-learn, TensorFlow, Keras, Scipy, NLTK, SpaCy
- **Other Technology:** Git, AWS, Microsoft Azure, Google Cloud, Linux, OpenCV, Hadoop, Kafka, Stata, Spark, Jupyter Notebook, Docker, Airflow
- **Other Skills:** REST APIs, Data Preprocessing, ETL, Generative-AI, Large Language Models, Statistics - Regression, T-test, Chi-squared, ANOVA

Work Experience

Utrecht It Consulting, Netherlands

Automation & Integration Consultant | *Automation and Integration, Machine Learning, API Integration & Cost Optimization*

Oct 2024 - Aug 2025

- Engineered custom SDK connectors leveraging Solace's pub/sub messaging model to decouple distributed data workflows
- Improving real-time event processing throughput by 30%
- Embedded custom AI agents using Workato's Agent Studio to orchestrate enterprise workflows with built-in governance and security
- Automated end-to-end operational workflows using Python, eliminating bottlenecks and enabling teams to ship reliable systems faster
- Designed fault-tolerant, scalable cloud integration solutions adopted across three production pipelines

Micropro Solutions, Nagpur India

Artificial Intelligence Engineering Intern | *Retrieval-Augmented Generation, Machine Learning, Data Analytics*

Mar 2024 - Sep 2024

- Deployed a multimodal RAG pipeline combining Whisper transcription and frame captioning to enable natural language video search
- Built query-time retrieval over Qdrant/FAISS with cross-encoder reranking, improving precision and relevance at scale
- Instrumented distributed AI services with OpenTelemetry and containerized the full stack via Docker Compose for production reliability
- Streamed live agent runs using Redis Pub/Sub and WebSockets, enabling real-time monitoring and deterministic offline replay

Research Paper

Reviewing Advances in Food Image Recognition and Nutritional Assessment: Focus on YOLOv8 | Oral Presentation in IEEE Journal

[GitHub](#)

Computer Science, Data Cleaning/Preprocessing, Data Augmentation, YOLOv8

- Co-authored a peer-reviewed research paper, benchmarking YOLOv8 against Faster R-CNN and SSD across speed and accuracy metrics
- Built and evaluated custom Indian food dataset with varied lighting and resolution, achieving 82% recognition on a 1,000-image Indian food benchmark
- Proposed an India-Food Dataset architecture with metadata on portion size and calorie content, addressing gaps in Food-101 and UEC-Food100
- Analyzed 3D volume estimation techniques including stereo vision, GANs, and SfM
- Identified scalable pathways for automating nutritional assessment in real-world dietary monitoring systems

Projects

VisionVault

Software Engineer | *Retrieval-Augmented Generation, Async Backend Architecture, Vector Search, API Integration & Cost Optimization*

- Developed a multimodal RAG retrieval pipeline combining speech transcription and optional vision cues aligned into searchable chunks
- Integrated LlamaIndex for optional RAG index construction and local LLM synthesis (Ollama) to enable "ask questions about the video" workflows
- Built query-time retrieval logic supporting multiple vector backends and tuned tradeoffs for speed vs accuracy via environment-configurable knobs
- Upgraded embedding quality and improved retrieval precision using cross-encoder reranking and dynamic result filtering for higher relevance

OrchestrAI

Software Engineer | *FastAPI, Postgres, Redis, Celery, Next.js, Docker*

- Built a local-first agent observability platform to record agent runs step-by-step with persistent storage in Postgres and migrations via Alembic
- Executed real-time run streaming using Redis Pub/Sub + WebSockets and added deterministic replay plus offline evaluation pipelines using Celery
- Instrumented services with OpenTelemetry, packaged the stack with Docker Compose and integrated both local and hosted LLM execution paths

CarWise-AI

Software Engineer | *AI Search Grounding, Real-Time Web Integration, React, TypeScript, Google Gemini API*

- Built a live-search powered car discovery assistant using Google Gemini's native Search Grounding to fetch real, verifiable listings in real-time
- Integrated Gemini Flash to auto-generate AI-driven pros, cons, and summaries for each listing, replacing hallucinated results with grounded web data
- Designed a side-by-side comparison grid surfacing mileage, price, location, and source links for data-driven purchase decisions
- Engineered graceful error handling for incomplete real-world listing data, ensuring reliable UX across inconsistent marketplace sources

Certification

Full Stack Web Development – Apna College([WDM](#))

Workato Automation Pro - I ([WAP-01](#))

Workato Automation Pro – II ([WAP-02](#))

Workato Automation Pro – III ([WAP-03](#))